Differentiate sigmoid Activation Function–

Here, sigmoid function $\alpha(x) = 1/1+e^{-x}$

chain rule Approach: $d/dx \left(1+e^{-x}\right)^{-1}$

$$= (-1)\left(1+e^{-x}\right)^{-2} \cdot d/dx\left(1+e^{-x}\right)$$

$$= (-1)\left(1+e^{-x}\right)^{-2} \cdot (-1) \cdot e^{-x}$$

$$= \left(1+e^{-x}\right)^{-2} \cdot e^{-x}$$

$$= \frac{e^{-x}}{\left(1+e^{-x}\right)^2}$$

$$\therefore \alpha'(x) = \frac{e^{-x}}{\left(1+e^{-x}\right)^2}$$

Here,

$$1-\alpha(x) = 1 - \cancel{\frac{e^{-x}}{\left(1+e^{-x}\right)^2}} \; \frac{1}{1+e^{-x}}$$

$$= \frac{1+e^{-x}-1}{1+e^{-x}} = \frac{e^{x}}{1+e^{-x}}$$

$$\therefore \alpha(x)\left(1-\alpha(x)\right) = \frac{1}{1+e^{-x}} \times \frac{e^{-x}}{1+e^{-x}}$$

$$= \frac{e^{-x}}{\left(1+e^{-x}\right)^2} = \alpha(x)$$

Therefore, $\alpha'(x) = \alpha(x) \cdot \left(1-\alpha(x)\right)$, (proved)